

HAVM: Designing 3D Hybrid-Anisotropic-Voronoi Cells for Microstructured Metamaterial

Haikuan Zhu
Wayne State University
Detroit, USA
hkzhu@wayne.edu

Sikai Zhong
Wayne State University
Detroit, USA
sikai.zhong@wayne.edu

Jing Hua
Wayne State University
Detroit, USA
jinghua@wayne.edu

Jimmy Ching-Ming Chen
Wayne State University
Detroit, USA
jcmchen@wayne.edu

Yongjie Jessica Zhang
Carnegie Mellon University
Pittsburgh, USA
jessicaz@andrew.cmu.edu

Charlie C.L. Wang
The University of Manchester
Manchester, United Kingdom
changling.wang@manchester.ac.uk

Zichun Zhong*
Computer Science
Wayne State University
Detroit, Michigan, USA
zichunzhong@wayne.edu

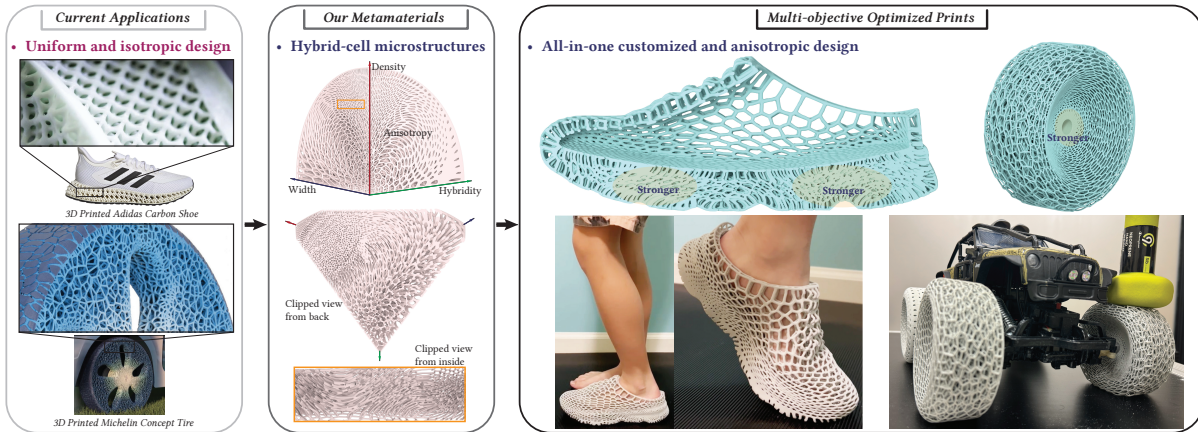


Figure 1: Left: Microstructures in cutting-edge applications (images are courtesy of Adidas and Michelin); Middle: Our geometric-controlled high-d parametric space of microstructured metamaterials; Right: Our modeling and printing results from a single base material (e.g., Shoe and Tire) satisfying the desired mechanical strength, material stiffness, and lightweight requirements.

Abstract

We introduce a novel class of hybrid-anisotropic microstructures governed by a set of fully geometric-driven parameters, enabling comprehensive top-down control for cell shape, orientation, density, hybridity (open and closed configurations), and thickness. To realize this, we develop a computational framework that transforms the designed parameter space into Voronoi-based micro-tessellations for

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771508>

3D hybrid-anisotropic cell structures. The numerical simulations and physical experiments are across multiple objectives, i.e., stiffness, strength, and weight. These results are obtained using only a single base material, demonstrating the versatility and applicability.

ACM Reference Format:

Haikuan Zhu, Sikai Zhong, Jing Hua, Jimmy Ching-Ming Chen, Yongjie Jessica Zhang, Charlie C.L. Wang, and Zichun Zhong. 2025. HAVM: Designing 3D Hybrid-Anisotropic-Voronoi Cells for Microstructured Metamaterial. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757374.3771508>

1 Introduction

The advent of additive manufacturing has enabled the creation of intricate volumetric microstructures, which can be engineered into metamaterials. However, recent works [Arora et al. 2019; Martinez

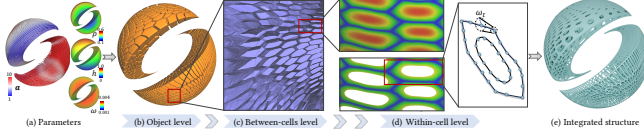


Figure 2: Modeling the multilevel metamaterial.

et al. 2018; Xu et al. 2023] typically offer limited control over the hierarchical parameters that are crucial to fully exploit the design space of metamaterials. Inspired by the flexible structure of human bone, which combines both plate-like and rod-like elements [Mathieu et al. 2006], we define the novel *hybridity* (openness / closedness) for each facet of cells, which are capable of continuously transitioning between open and closed configurations. Moreover, *anisotropy* allows the microstructures to better accommodate uneven mechanical properties. For instance, the stiffness and strength of wood are much higher along the long axis of the cells [Gibson 2005]. However, this aspect has been little explored, which offer limited degrees of freedom to the 2D plane only [Li et al. 2023; Martínez et al. 2018, 2017]. To address this, we introduce cell-architected parameters to achieve extended *anisotropy*. At the cell level, we enable fully 3D anisotropy by aligning the structure with specified 3D tensor fields (e.g., stress tensors). At the within-cell level, we propose orientation-associated parameters—hybridity and thickness—to introduce additional anisotropy. These parameters can both expand the accessible material space and enhance structural strength. This work proposes a new computational pipeline to generate the versatile 3D hybrid-anisotropic-Voronoi volumetric micro-tessellations for designing microstructured metamaterials.

2 Method

Geometric Design of HAVM. We propose fully geometric-controlled parameters for HAVM and develop a computational pipeline to generate hybrid-anisotropic Voronoi-based microstructures as shown in Fig. 2. Given a normalized 3D object Ω represented as a tetrahedral mesh, we define a set of parameters w.r.t. cell’s anisotropy and hybridity to each mesh vertex $t \in \Omega$ to control cell shape, size, shell, and beam from the output microstructures as follows:

- (1) **Cell’s Anisotropy and Density (Between-Cells Level):**
 - **Anisotropy $\mathbf{a}$,** defined as three normalized orthogonal directions $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ and their magnitudes $\{r_1, r_2, r_3\}$, controls the shape of cells, where $\mathbf{a}_3 = \mathbf{a}_1 \times \mathbf{a}_2$ and $r_1 \geq r_2 \geq r_3$. The local orthogonal axes can be oriented in arbitrary directions.
 - **Density ρ ,** defined as the number of cells in a unit space, allows the spatially graded distribution of cells. We use a symmetric matrix $\mathbf{A}$ to represent the cell anisotropy and density in a unified way: $\mathbf{A} = \rho^2 \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}^T \text{Diag}(r_1^2, r_2^2, r_3^2) \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$.
- (2) **Cell’s Hybridity (Openness $\leftrightarrow$ Closedness) and Thickness (Within-Cell Level):**
 - **Hybridity h ,** defined as the orientation-associated closed ratio of a cell shell, which can be considered as a hole in a curved patch (cell facet), controls the transition in-between the cells. $h = 0$ or 1 indicates an open or closed shell.
 - **Thickness ω ,** defined as the orientation-associated radius along the beam nodes, controls the thickness of shells and supports the local anisotropy. The

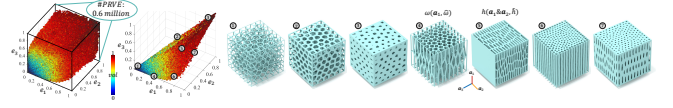


Figure 3: Our HAVM material space. Left: Full sampling space; Middle: One of six sub-spaces; Right: Several PRVE samples (colormap denotes the volume) are illustrated.

orientation attributes of hybridity h and thickness ω provide additional anisotropy at the fine level, which is highly effective for stiffness and strength design. Given a user-specified direction $\mathbf{d}_f$, hybridity h becomes larger if the shell normal $\mathbf{d}_s$ is perpendicular with $\mathbf{d}_f$. Thickness ω becomes larger if its beam’s direction $\mathbf{d}_b$ is aligned with $\mathbf{d}_f$. So, we can define: $h(\mathbf{d}_f, \hat{h}, h_{min}) = (1 - |\text{dot}(\mathbf{d}_s, \mathbf{d}_f)|)(\hat{h} - h_{min}) + h_{min}$, $\omega(\mathbf{d}_f, \hat{\omega}, \omega_{min}) = |\text{dot}(\mathbf{d}_b, \mathbf{d}_f)|(\hat{\omega} - \omega_{min}) + \omega_{min}$, where $\hat{h} \geq h_{min}$ and $\hat{\omega} \geq \omega_{min}$. $h_{min} = 0$ and $\omega_{min} = 0.008$ can be omitted.

Material Behavior of HAVM. We analyze the mechanical behavior of our HAVM in terms of stiffness and strength. From our observations and analyses, stretching ratios and orientation-related hybridity / thickness promote strength with increasing minimal volume cost and are keys to the anisotropy of stiffness. In contrast, density and hybridity / thickness (without orientation) also play a key role by controlling the consumed volume in an isotropic or adaptive way. This is another factor influencing both stiffness and strength. The purpose of lightweight HAVM is to simultaneously fulfill the anisotropic stiffness $\mathbb{E}_{input}$ while enhancing the strength.

We first build a quantitative relationship between elastic properties and our geometric-controlled parameters. We uniformly sample our HAVM($\mathbf{a}, \rho, h, \omega$) in a parametric representative volume elements (PRVE) with 0.6 million samples as shown in Fig. 3. The 4th and 5th PRVE samples exhibit the local anisotropy of orientation-associated thickness / hybridity. The homogenized stiffness tensor of each PRVE pv is determined utilizing the Fibergen library. We fit the homogenized stiffness tensor to an ideal orthotropic tensor, which retrieves Young’s moduli. We focus on Young’s modulus (e_1, e_2, e_3) to analyze the stiffness for each orthotropic PRVE pv . HAVM strength $\mathbb{S}_{HAVM}$ is a scalar stress field based on Von Mises values ξ from stress analysis. A local element starts yielding when its von Mises stress ξ reaches a constant value known as yield strength χ determined by the base material. In this case, the HAVM strength is improved by reducing the peak stress ξ_{max} .

3 Results and Applications

We consider strength, stiffness, and lightweight simultaneously in designing applications in Fig. 1. The first example is an all-in-one customized running shoe. This is achieved by generating a flexible top and improving the strength of the bottom, ensuring an enduring, comfortable, and lightweight shoe. This shoe demonstrates the bending effects at the upper part and its capacity to withstand a 70lbs kid. Moreover, we demonstrate two different types of airless tires, whose fabricated diameter is around 100mm—stiffer one and more flexible one—weighing 148g and 99g, respectively. To satisfy

the need for strength in the inner part, we align the anisotropic orientation along the tire's axial direction. Conversely, to maintain flexibility in the outer part, we orient the anisotropy vertically to the axial direction. When subjected to an equivalent load of 5lbs, they demonstrate normalized deformation displacements of 10% and 25%, respectively.

References

- Rahul Arora, Alec Jacobson, Timothy R Langlois, and et al. 2019. Volumetric Michell trusses for parametric design & fabrication. In *Proceedings of the ACM Symposium on Computational Fabrication*. 1–13.
- Lorna J Gibson. 2005. Biomechanics of cellular solids. *Journal of Biomechanics* 38, 3 (2005), 377–399.
- Yue Li, Stelian Coros, and Bernhard Thomaszewski. 2023. Neural metamaterial networks for nonlinear material design. *ACM Transactions on Graphics* 42, 6 (2023), 1–13.
- Jonàs Martínez, Samuel Hornus, Haichuan Song, and Sylvain Lefebvre. 2018. Polyhedral Voronoi diagrams for additive manufacturing. *ACM Transactions on Graphics* 37, 4, Article 129 (2018).
- Jonàs Martínez, Haichuan Song, Jérémie Dumas, and Sylvain Lefebvre. 2017. Orthotropic K-nearest foams for additive manufacturing. *ACM Transactions on Graphics* 36, 4, Article 121 (2017).
- Laurence Marcelle Mathieu, Thomas L. Mueller, Pierre-Etienne Bourban, and et al. 2006. Architecture and properties of anisotropic polymer composite scaffolds for bone tissue engineering. *Biomaterials* 27, 6 (2006), 905–916.
- Wenpeng Xu, Peng Zhang, Menglin Yu, Li Yang, Weiming Wang, and Ligang Liu. 2023. Topology optimization via spatially-varying TPMS. *IEEE Transactions on Visualization and Computer Graphics* 30, 8 (2023), 1–18.